Blessed G/Michael Catholic Secondary School

Semester 1 Biology Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Dichotomous keys are useful for taxonomists to classify unfamiliar organisms.

____ **2.** Biology is related to chemistry and physics during studying the process of photosynthesis.

____ **3.** The correct way of handling/carrying a microscope is holding it with one hand only.

____ **4.** It is permissible to taste and sniff chemicals in laboratory rooms.

____ **5.** In symbiotic association between fungi and termites, fungi supply extracellular hydrolytic enzymes to break down cellulose.

II. Match the items under Column A with the correct descriptions under Column B.

30. Which pair correctly matches the organism with its cell wall chemical composition?

A. Cyanobacteria — Peptidoglycan cell wall
B. Cyanophyta — Cellulose cell wall
C. Yeast — Chitin & glucan cell wall
D. All of the above are correct

III. Choose the best answer among the given alternatives.

____ **6.** Which part of the optical microscope is used to move the mechanical stage rapidly for initial focusing?

A. Fine adjustment knob **B.** Coarse adjustment knob
C. Condenser **D.** Diaphragm

____ **7.** Which part of the microscope moves the stage slowly to produce a sharp, crisp focus?

A. Fine adjustment knob **B.** Coarse adjustment knob
C. Ocular lens **D.** Stage clips

____ **8.** Which component of a microscope controls the intensity and amount of light reaching the specimen?

A. Diaphragm **B.** Condenser **C.** Coarse knob **D.** Nosepiece

____ **9.** Which microscope part collects and focuses light rays from the illuminator onto the specimen?

A. Condenser **B.** Diaphragm **C.** Objective lens **D.** Mirror

____ **10.** Which apparatus is used in laboratories to provide high-pressure steam for complete sterilization?

A. Autoclave **B.** Pressure cooker
C. Incubator **D.** Hot oven

____ **11.** The ancient Greek philosopher who divided animals into two groups (those with red blood and those without) was:

A. Whittaker **B.** T.O. Diener **C.** Aristotle **D.** W.M. Stanley

____ **12.** The criteria used in R.H. Whittaker's five-kingdom classification system include:

A. Cell structure (prokaryotic vs. eukaryotic)
B. Mode of nutrition and reproduction
C. Phylogenetic evolutionary relationships
D. All of these

____ **13.** Which of the following biological groups are included within Whittaker's five kingdoms?

A. Viruses **B.** Bacteria
C. Lichens **D.** Bacteria and Algae

____ **14.** The naturalist who developed binomial nomenclature and authored "Systema Naturae" is:

A. Carolus Linnaeus **B.** Aristotle
C. Charles Darwin **D.** Jean-Baptiste Lamarck

____ **15.** What primary method of asexual reproduction is used by yeast?

A. Sexual spore formation **B.** Budding

C. Regeneration **D.** Conjugation

____ **16.** Protists are ecologically vital because they:

A. Produce oxygen via phytoplankton photosynthesis
B. Support marine and freshwater food chains
C. Recycle organic nutrients in aquatic habitats
D. All of the above

____ **17.** Fungi are indispensable to global ecosystems primarily because they function as:

A. Producers **B.** Regulators

C. Decomposers (saprotrophs) **D.** Controllers

____ **18.** Which statement is true regarding a well-designed controlled experiment?

A. Demonstrates cause-and-effect relationships
B. Isolates one independent variable while measuring a dependent variable
C. Keeps all other environmental variables strictly controlled
D. All are true expressions

____ **19.** Which of the following is an example of a non-hyphal (unicellular) fungus?

A. Yeast **B.** Mucor **C.** Rhizopus **D.** Mushroom

____ **20.** Which of the following cellular structures are used as locomotive appendages in unicellular organisms?

A. Flagella **B.** Cilia

C. Pseudopodia **D.** All of the above

____ **21.** Which of the following protists is free-living and does NOT cause human infectious disease?

A. Amoeba (Entamoeba) **B.** Paramecium

C. Plasmodium **D.** Giardia

____ **22.** Which method of asexual reproduction is characteristic of Paramecium?

A. Budding **B.** Sporulation

C. Binary fission **D.** Multiple fission

____ **23.** Which of the following is NOT a difference between fungi and animals?

A. Fungi have chitinous cell walls whereas animals lack cell walls
B. Fungi are heterotrophs while animals are autotrophs
C. Fungi are absorptive heterotrophs while animals are ingestive heterotrophs
D. Fungi are non-motile while animals are motile

____ **24.** Fungi resemble animals because they are:

A. Heterotrophs (store glycogen, non-photosynthetic)
B. Heterosporous
C. Chemoautotrophs
D. Photoautotrophs

____ **25.** All of the following are asexual modes of reproduction in fungi EXCEPT:

A. Budding **B.** Sporulation **C.** Fragmentation **D.** Conjugation

____ **26.** The vegetative body of multicellular fungi (mycelium) is composed of microscopic tubular branched filaments called:

A. Hyphae **B.** Sporangium **C.** Rhizoids **D.** Spores

____ **27.** Protozoan protists are traditionally classified into classes primarily according to:

A. How they obtain nutrients
B. How they grow
C. Their mechanism of locomotion (cilia, flagella, pseudopodia)
D. How they reproduce

____ **28.** Which of the following Archaebacteria can thrive in aerobic, hypersaline environments?

A. Halophiles **B.** Methanogens

C. Thermoacidophiles **D.** Cyanobacteria

___ **29.** During binary fission, which statement accurately describes the resulting daughter cells?

A. They have half as many chromosomes as the parent cell
B. They have double the chromosome number
C. They have the same chromosome count as the parent but initial smaller cell volume
D. They contain no genetic material